

CS5590 Foundations of Machine Learning

Instructor: J. SakethaNath

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About the course

This is the basic foundations course on machine learning for CSE students. The course assumes no prior exposure to machine learning. However, it assumes the student has good understanding of basic engineering math subjects like probability & statistics, linear algebra, multivariate Calculus etc.

The focus of this course is on understanding basic principles and concepts, rather than on giving a hands-on overview of state-of-the-art. Students interested in the later are encouraged to credit the “Applied ML” course.

The syllabus of this course is:

1. Overview of machine learning, challenges, terminology and definitions, Supervised Learning (5 hrs)
2. Basic non-probabilistic models:
 - (a) Loss composed Linear Models and parameter estimation using ERM&SGD (5 hrs)
3. Generalized models:
 - (a) Exponential-family models, and kernel methods (3 hrs)
 - (b) Neural Networks and deep models (4 hrs)
4. Model selection (1 hrs)
5. Unsupervised Learning
 - (a) Likelihood estimation (3hrs)
 - (b) Sampling (GenAI models) (3hrs)
6. Basic probabilistic models for Supervised Learning:
 - (a) Generative models and parameter estimation using method of moments, maximum likelihood (1hrs)
 - (b) Discriminative models and maximum conditional likelihood estimation (2 hrs)

7. Non-parametric modeling (2 hrs)
 - (a) k-NN, KDE
 - (b) variants for generative, discriminative classification and regression
8. Hidden Variables: Mixture models, EM algorithm, k-means (3hrs)
9. Representation Learning (5 hrs)
 - (a) PCA, probabilistic PCA, supervised PCA
 - (b) Encoder-Decoder architectures
10. Online Learning (2 hrs)
11. Reinforcement Learning: Policy gradient (4 hrs)
12. Other popular paradigms: Boosting, Bagging, trees (3 hrs)

References

The textbooks for this course are:

1. For probabilistic models and neural networks [1]¹.
2. For kernel methods [2].
3. In general, part2 and part 3 of [3]².

Evaluation

Date	Name & Duration	Percentage
06-Sep-2025 (Sat)	Quiz-1 7pm-8:30pm (1.5hrs.)	15% (Closed book/laptop; No cheat-sheets :)
25-Oct-2025 (Sat)	Quiz-2 7pm-8:30pm (1.5hrs.)	15% (Closed book/laptop; No cheat-sheets)
22-Nov-2025 (Sat)	EndSem 9:30am-12:30pm (3hrs.)	55% (Closed book/laptop; No cheat-sheets)
23-Nov-2025 (Sun)	Coding Exam 9:30am-11am (1.5hrs.)	15% (Closed book/laptop; No cheat-sheets)
Biweekly	Tutorials (8)	Scope for negative marks(-2% per session)

Tutorial sessions will be conducted for help in practice problems.

¹Available at <https://probml.github.io/pml-book/book1.html>.

²This is freely available at <https://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/understanding-machine-learning-theory-algorithms.pdf>.

Contact

The course page is at <http://www.iith.ac.in/saketha/teaching/cs5590.html>. All official communication and technical interactions outside lecture hours will be through Google classroom: “xcuye6qc”. The instructor can be contacted via email: saketha at cse.iith. The typical working hours of the instructor are 9am to 1pm and 2:30pm to 5:30pm. His office is CS-510 (EECS Building). You are welcome to drop by anytime to clarify doubts during these working hours (ping me on chat to confirm if I have not stepped out etc.). Technical interaction (outside and in lecture hours) is highly encouraged.

Apart from this there will be a dedicated 2-3hr interactive tutorial session conducted bi-weekly. This can be used to clear up doubts in weekly practice problems. If time permits, it can also be used to clarify lecture doubts. Finally, TAs will use it to conduct sessions related to coding/ML-libraries, useful in coding exam.

References

- [1] Kevin P. Murphy. *Probabilistic Machine Learning: An introduction*. MIT Press, 2022.
- [2] Bernhard Scholkopf and Alexander J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, Cambridge, MA, USA, 2001.
- [3] Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, New York, NY, USA, 2014.